

University of Cape Town
Department of Mathematics and Applied Mathematics

2IA Class Test 1.5
26th September, 2023
Time: 1,5 hours
Total marks: 105
Full marks: 100

- Read the instructions carefully.
- You have 1,5hrs to complete the test.
- Make sure that no pages or questions are missing.
- Show all your work or reasoning. Answers without work or reasoning shown will receive no mark.

Surname: _____

First name: _____

Student number: _____

Signature: _____

Do not write below this line:

Run L^AT_EX again to produce the table

1. (a) 5 marks State Bézout's Lemma.
- (b) Consider the following incomplete proof of the **Bézout's Lemma**:

Let $X = \{sm + tn \mid s, t \in \mathbb{Z} \text{ and (A) } \}$. Note that X is non-empty, since $m^2 + n^2 \in X$. By the (B) we can find a least element in X , say d . Then $d \geq 1$ and $d = xm + yn$ for integers x and y ; if $k|m$ and $k|n$, then $k|(xm + yn) = d$ by a theorem proved in class (Theorem 1.10 (4) to be more precise). This shows that any common divisor of m and n is a divisor of d . It remains to show that d is a common divisor of m and n . Here, we will show that $d|n$. Similarly, one can show that $d|m$. Apply the (C1) to find $q, r \in \mathbb{Z}$ such that (C2) and (C3). Then

(D)

With reference to this partial proof:

- i. 2 marks Complete the sentence at (A).
- ii. 2 marks Complete the sentences at (B).
- iii. 6 marks Complete the sentence at (C1), (C2) and (C3).
- iv. 10 marks Fill in the necessary steps at (D) to conclude that $r = 0$.
- (c) 10 marks Let a, b be integers such that $\gcd(a, b) = 1$. Use the Principle of Mathematical Induction to prove that

$$\gcd(a, b^n) = 1 \text{ for all } n \geq 1.$$

Solution: (a) Bézout's Lemma: Let m and n be two integers not both zero. Then $d = \gcd(m, n)$ exists and $d = xm + yn$ for some integers x and y .

(b) i. $sm + tn \geq 1$.

ii. Well-Ordering Axiom.

iii. Division Algorithm; $n = qd + r$ and $0 \leq r < d$.

iv. $r = n - qd = n - q(xm + yn) = (-qx)m + (1 - qy)n$, which says that r is a linear combination of m and n . If $r > 0$ we would have that $r \in X$, which contradicts the minimality of d in X . Thus $r = 0$.

(c) Suppose that a, b are integers such that $\gcd(a, b) = 1$. For $n \geq 1$, consider the statement:

$$p_n: \quad \gcd(a, b^n) = 1$$

Notice that p_1 is true by hypothesis. Assume that p_k is true for some $k \geq 1$ (that is, $\gcd(a, b^k) = 1$) and let us show that p_{k+1} is also true. We need to prove that $\gcd(a, b^{k+1}) = 1$. From Bézout's Lemma we get that

$$ax + by = 1 \text{ and } ax' + b^k y' = 1,$$

for some integers x, y, x', y' . From here we get that

$$\begin{aligned} 1 &= 1 \cdot 1 = (ax + by)(ax' + b^k y') = aax' + ab^k xy' + abx'y + bb^k yy' \\ &= a(ax' + b^k xy' + bx'y) + b^{k+1} yy'. \end{aligned}$$

Since the converse of Bézout's Lemma holds for $d = 1$, from here we can conclude that $\gcd(a, b^{k+1}) = 1$, proving that p_{k+1} is true.

2. In each case determine whether the statement is true or false. **You will get zero marks if you do not provide an argument supporting your answer.**

- (a) 6 marks Suppose that n and m are two integers not both zero, and $d > 1$ is an integer satisfying that $d = xm + yn$ for some integers x and y . Then $d = \gcd(n, m)$.
- (b) 6 marks Let a, b, n, m be integers with $n \geq 2$ and $m \geq 2$. If $\bar{a} = \bar{b}$ in $\mathbb{Z}_{nm}$, then $\bar{a} = \bar{b}$ in $\mathbb{Z}_n$.
- (c) 6 marks $2 + 70 \equiv 3^3 \pmod{5}$
- (d) 6 marks There exist unique integers x and y such that $1 = 7x + 11y$.
- (e) 7 marks If $a \in \mathbb{Z}$, then $\bar{a}(\bar{a} + \bar{1})(\bar{a} + \bar{2}) = \bar{0}$ in $\mathbb{Z}_6$.
- (f) 6 marks If $\sigma, \tau, \mu \in S_n$ are such that $\sigma\tau = \mu\sigma$, then $\tau = \mu$.
- (g) 6 marks If σ and τ fix k , then $\sigma\tau$ also fixes k .

Solution: (a) **False:** For example: $12 = 1 \cdot 4 + 1 \cdot 8$, and 12 is not the gcd of 4 and 8, since $\gcd(4, 8) = 4$.

(b) **True:** From $\bar{a} = \bar{b}$ in $\mathbb{Z}_{nm}$ we get that $a \equiv b \pmod{nm}$. This means that $nm \mid (a - b)$; that is, $a - b = (nm)k = n(mk)$ for some $k \in \mathbb{Z}$. But then $a - b$ is also a multiple of n , and so $a \equiv b \pmod{n}$, which is equivalent to say that $\bar{a} = \bar{b}$ in $\mathbb{Z}_n$.

(c) **True:** $(2 + 70) - 3^3 = 72 - 27 = 45$ and $5 \mid 45$.

(d) **False:** The integers x and y do not need to be unique. For instance, applying the Euclidean Algorithm (notice that $\gcd(11, 7) = 1$) we get

$$\begin{aligned} 11 &= 1 \cdot 7 + 4 \Rightarrow 4 = 11 + (-1)7, \\ 7 &= 1 \cdot 4 + 3 \Rightarrow 3 = 7 + (-1)4, \\ 4 &= 1 \cdot 3 + 1 \Rightarrow 1 = 4 + (-1)3. \end{aligned}$$

From here, we get that

$$\begin{aligned} 1 &= 4 + (-1)3 = 4 + (-1)(7 + (-1)4) = (-1)7 + 2 \cdot 4 \\ &= (-1)7 + 2 \cdot (11 + (-1)7) = 2 \cdot 11 + (-3)7 \end{aligned}$$

But also, $1 = (-5)11 + 8 \cdot 7$.

(e) **True:** Recall $\mathbb{Z}_6 = \{\bar{0}, \bar{1}, \bar{2}, \bar{3}, \bar{4}, \bar{5}\}$. For any $a \in \mathbb{Z}$, we proved in class that $\bar{a} = \bar{r}$, where $r \in \{0, \dots, 5\}$ is the remainder obtained when a is divided by 6. It is enough to check that the equation $\bar{a}(\bar{a} + \bar{1})(\bar{a} + \bar{2}) = \bar{0}$ is satisfied for $\bar{r}$, where $r \in \{0, \dots, 5\}$:

- for $r = 0$, the equation clearly holds;

- for $r = 1$, we have that $\bar{1}(\bar{1} + \bar{1})(\bar{1} + \bar{2}) = \bar{1} \cdot \bar{2} \cdot \bar{3} = \bar{6} = \bar{0}$;
- for $r = 2$, we get that $\bar{2}(\bar{2} + \bar{1})(\bar{2} + \bar{2}) = \bar{2} \cdot \bar{3} \cdot \bar{4} = \overline{24} = \bar{0}$;
- for $r = 3$, we have that $\bar{3}(\bar{3} + \bar{1})(\bar{3} + \bar{2}) = \bar{3} \cdot \bar{4} \cdot \bar{5} = \overline{60} = \bar{0}$;
- for $r = 4$, we obtain that $\bar{4}(\bar{4} + \bar{1})(\bar{4} + \bar{2}) = \bar{4} \cdot \bar{5} \cdot \bar{6} = \overline{120} = \bar{0}$;
- for $r = 5$, we get that $\bar{5}(\bar{5} + \bar{1})(\bar{5} + \bar{2}) = \bar{5} \cdot \bar{6} \cdot \bar{7} = \overline{210} = \bar{0}$.

(f) **False:** Take, for example, $\sigma = (1 \ 2)$, $\tau = (1 \ 2 \ 3)$ and $\mu = (1 \ 3 \ 2)$ (Here you can use matrix notation, if you feel more comfortable with it). In this case, we have that $\sigma\tau == \mu\sigma$, and clearly

(g) **True:** Suppose that σ and τ fix k , that is, $\sigma(k) = \tau(k) = k$. Then:

$$(\sigma\tau)(k) = \sigma(\tau(k)) = \sigma(k) = k,$$

which shows that $\sigma\tau$ fix k .

3. (a) 10 marks Find **all the solutions** to the simultaneous congruences:

$$x \equiv 8 \pmod{10} \quad \text{and} \quad x \equiv 3 \pmod{9}.$$

- (b) 7 marks Use (a) to find a solution (this mean one solution) of the simultaneous congruences given below. **If you give us a solution with no method or proof, you will receive 0 marks.**

$$x \equiv 8 \pmod{10}, \quad x \equiv 3 \pmod{9} \quad \text{and} \quad x \equiv 2 \pmod{7}.$$

Solution: (a) Since $\gcd(10, 9) = 1$, an application of the Classical Chinese Remainder Theorem tells us that there is a common solution. Moreover, any two solutions are congruent modulo $9 \cdot 10 = 90$.

We proceed like in the proof of the Classical Chinese Remainder Theorem. Any solution of $x \equiv 8 \pmod{10}$ is of the form $x = 8 + 10k$, for $k \in \mathbb{Z}$. Substituting in the first congruence we obtain:

$$8 + 10k \equiv 3 \pmod{9} \Leftrightarrow 10k \equiv -5 \pmod{9}$$

Next, since $10 \equiv 1 \pmod{9}$ and $-5 \equiv 4 \pmod{9}$ our equation becomes:

$$k \equiv 4 \pmod{9}$$

Taking (for example) $k = 4$, we obtain that a common solution of the simultaneous congruences is $x = 8 + 10 \cdot 4 = 48$; by the Chinese Theorem Remainder we can conclude that all the solutions are:

$$48 + (9 \cdot 10)n = 48 + 90n, \quad n \in \mathbb{Z}.$$

(b) From (a) we know that any solution of the first two congruences is of the form $48 + 90n$ for $n \in \mathbb{Z}$. Our goal is to find n such that $48 + 90n$ is a solution of the remaining congruence equation $x \equiv 2 \pmod{7}$. To do so, we substitute in x and solve the resulting equation for n :

$$48 + 90n \equiv 2 \pmod{7} \Rightarrow 90n \equiv -46 \pmod{7}$$

Now notice that $90 \equiv 6 \pmod{7}$ and $-46 \equiv 3 \pmod{7}$. Thus, the equation above becomes $6n \equiv 3 \pmod{7}$. Using that $\bar{6}$ is self-inverse in $\mathbb{Z}_7$ we obtain that $n \equiv 18 \pmod{7}$. Taking, for example, $n = 18$ we can conclude that $48 + 90 \cdot 18 = 1668$ is a solution of three congruences.

4. 10 marks Let n be an integer such that $n \geq 2$. Prove that the following assertions are equivalent.

You need to prove that (a) implies (b), and that (b) implies (a).

- (a) If $\bar{a} \in \mathbb{Z}_n$, then either $\bar{a}$ has a (multiplicative) inverse in $\mathbb{Z}_n$, or $\bar{a}^k = \bar{0}$ for some $k \geq 1$.
- (b) n is a power of a prime.

Solution: (a) implies (b):

Assume that any element $\bar{a} \in \mathbb{Z}_n$ has a multiplicative inverse, or $\bar{a}^k = \bar{0}$ for some $k \geq 1$.

Suppose, on the contrary, that (b) does not hold. Then n is of the form $n = p^r \cdot t$, where $t \in \mathbb{Z}$ is such that p does not divide t . Consider the element $\bar{t} \in \mathbb{Z}_n$. Notice that $\gcd(t, n) \neq 1$, and so $\bar{t}$ does not have a multiplicative inverse in $\mathbb{Z}_n$. If $\bar{t}^k = \bar{0}$ for some $k \geq 1$, then $n | t^k$. Since $p | n$ and $n | t^k$, we get that $p | t$ (by Euclid's Lemma), a contradiction.

We have found an element $\bar{a} \in \mathbb{Z}_n$ that does not satisfy any of the two conditions given in (a). This contradicts that (a) holds.

(b) implies (a): Suppose that $n = p^s$ for some prime p , and let $\bar{a} \in \mathbb{Z}_n$. If p does not divide a , then $\gcd(a, n) = 1$, which implies that $\bar{a}$ has a multiplicative inverse in $\mathbb{Z}_n$. If $p | a$, then $a = pq$ for some $q \in \mathbb{Z}$. But then

$$\bar{a}^s = \overline{pq}^s = \bar{p}^s \cdot \bar{q}^s = \bar{n} \cdot \bar{q}^s = \bar{0},$$

since $n = p^s$. This proves that (a) holds.